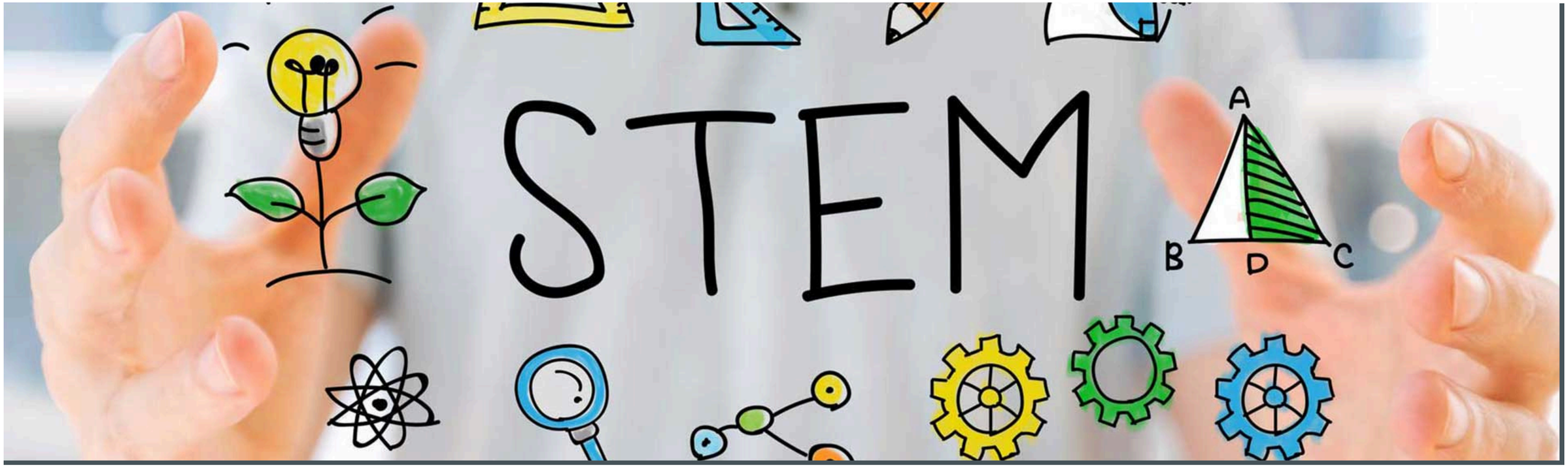
Study Buddy

STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

Project ID : R26-IT-009



PP2 Comments and Actions Taken

Topic	Early STEM learning through AR storytelling			
Project ID	R26-IT-009			
Specialization	IT			
Student ID	IT22087638	IT22136756	IT22272768	IT22292568
PanelName	IT-P01			
Detailed Comments	* Needs to have a clear idea on why the models have selected, and on what considerations, parameters used, why? and need to use fine-tuning techniques and a do model-comparison. * Look for fine-tuning techniques. * Plan to get real-user feedback from target users.	* Needs to have a clear idea on why the models have selected, and on what considerations, parameters used, why? and need to use fine-tuning techniques and a model-comparison. * Look for fine-tuning techniques.	* No proper idea about the architectures used. * Please review the logic used for the question generation in each level. * Needs to have a clear idea on why the models have selected, and on what considerations, parameters used, why? and need to use fine-tuning techniques and do a model-comparison.	* Needs to do a model comparison and have a clear idea on why the models have selected, and on what considerations, parameters used, why?
reponse		Selected Qwen2.5 7B because it gave the best overall balance of answer quality and speed. to fine-tuned it using SFT and QLoRA, then compare the original and improved models using the same questions and evaluation criteria.		

Problem and Design Excellence

Problems Addressed

- Low student engagement in STEM subjects.
- Difficulty understanding abstract scientific concepts.
- Limited access to personalized tutoring and revision assistance.
- Assessment stress and poor knowledge retention.
- Lack of integrated multilingual and multimodal learning support.

Our Solution: Study Buddy

Study Buddy integrates AI-powered educational capabilities into a single learning ecosystem.

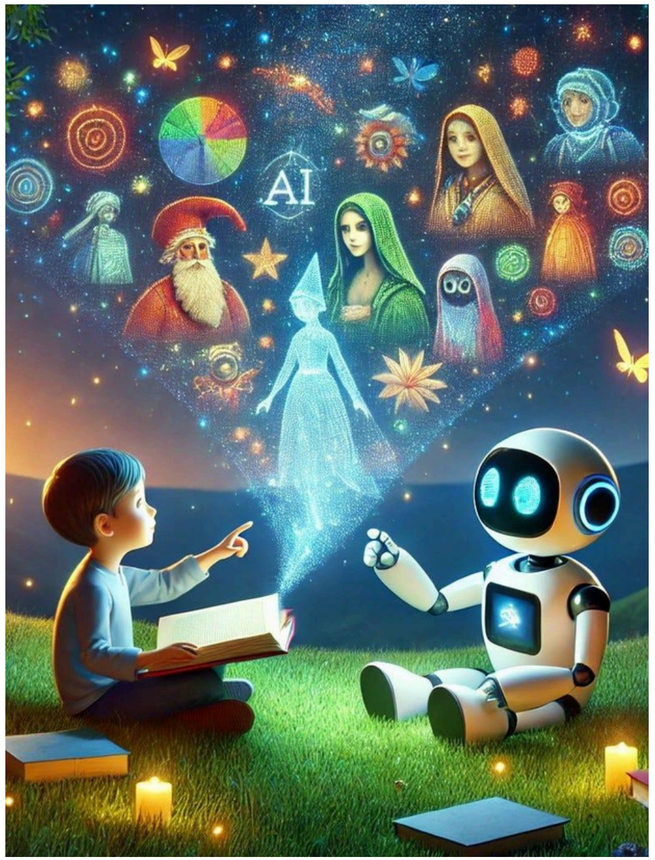
Key Design Features

- Personalized learning
- Conversational STEM tutoring
- Adaptive assessment
- AI-generated narratives
- Automated study materials
- Syllabus-grounded responses

Design Excellence

Rather than treating tutoring, assessment and study-material generation as separate activities, Study Buddy aims to integrate them into one personalized and adaptive STEM learning experience.

RAG-Based Personalized STEM Storytelling



Develops interactive and personalized STEM stories using Retrieval-Augmented Generation (RAG) to ensure accuracy, syllabus alignment and real-world relevance. It helps students understand complex concepts through engaging narratives and real-life scenarios.



IT22087638
ELWITIGALA S.A.I.

Study Buddy



Multilingual Conversational AI for STEM Tutoring

Provides context-aware tutoring through text, voice and image inputs in multiple languages. It supports concept explanation, doubt solving and step-by-step learning guidance.



IT22136756
MANODYA K .M. B.

Study Buddy



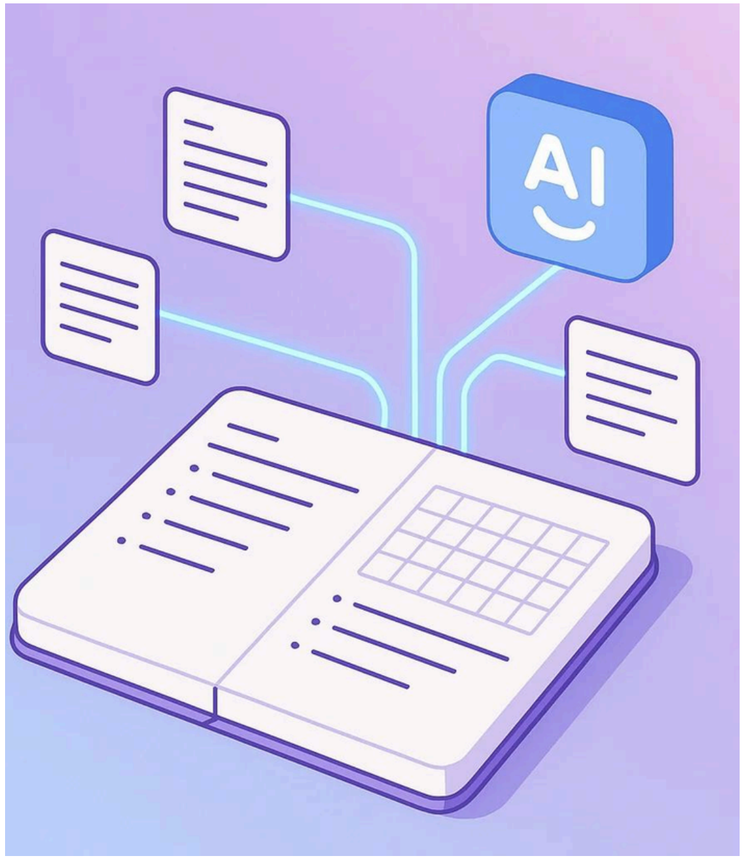
Emotion-Aware Adaptive Assessment

Detects student emotions and learning behaviour to adapt question difficulty, provide personalized feedback and recommend suitable learning resources. It aims to improve engagement and reduce assessment stress.



IT22272768
BASNAYAKA W.B. M.D.K.

Study Buddy



Handwritten Content Analysis & Knowledge Synthesis

Extracts handwritten notes using OCR and generates mind maps, flashcards, summaries and concept verification. It helps students revise better and convert notes into structured, easy-to-learn knowledge.

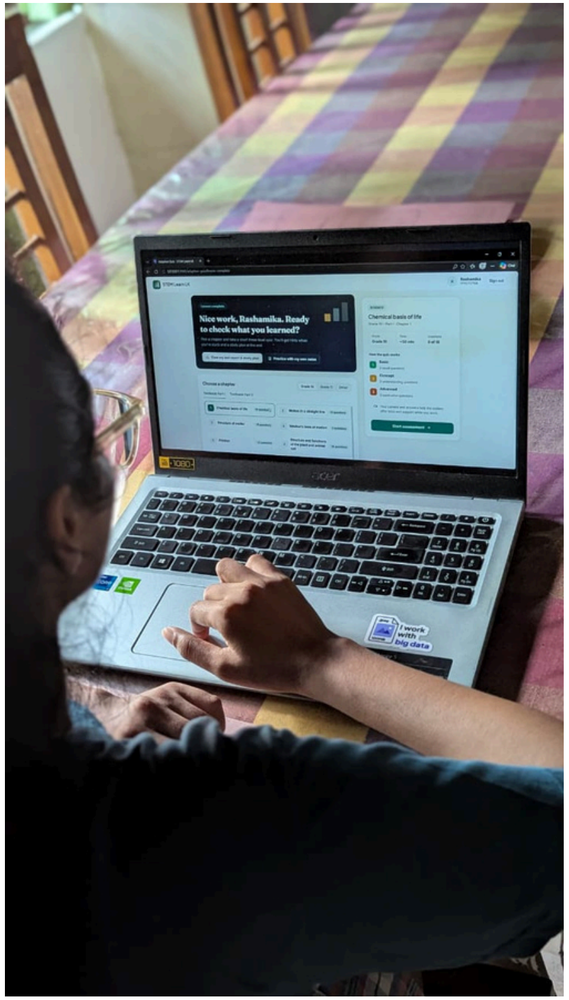


IT22292568
DILSHAN W.A.

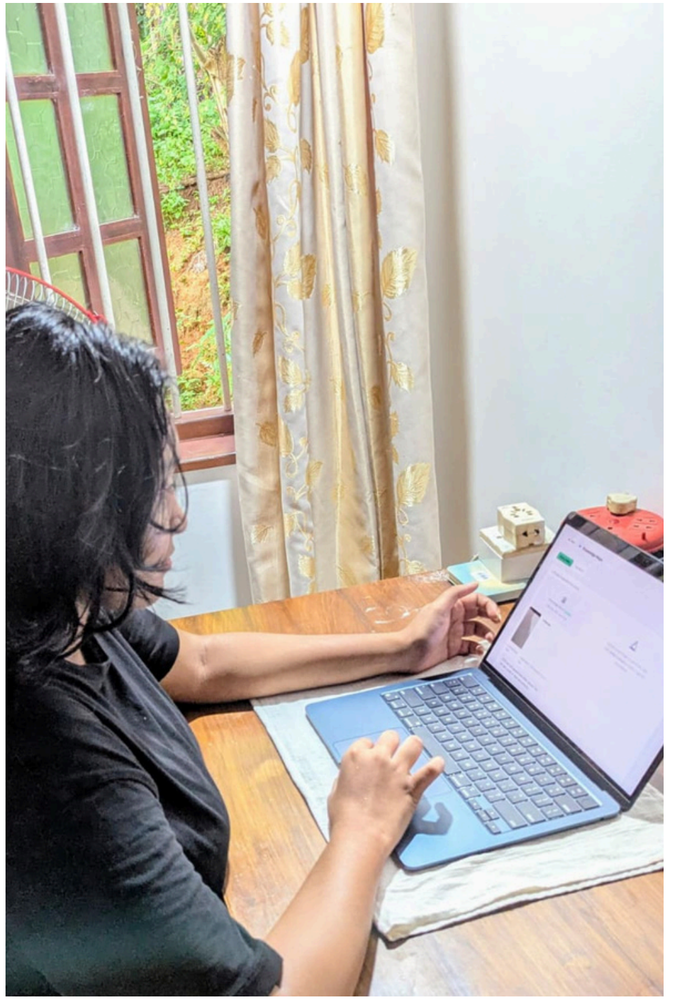
Study Buddy

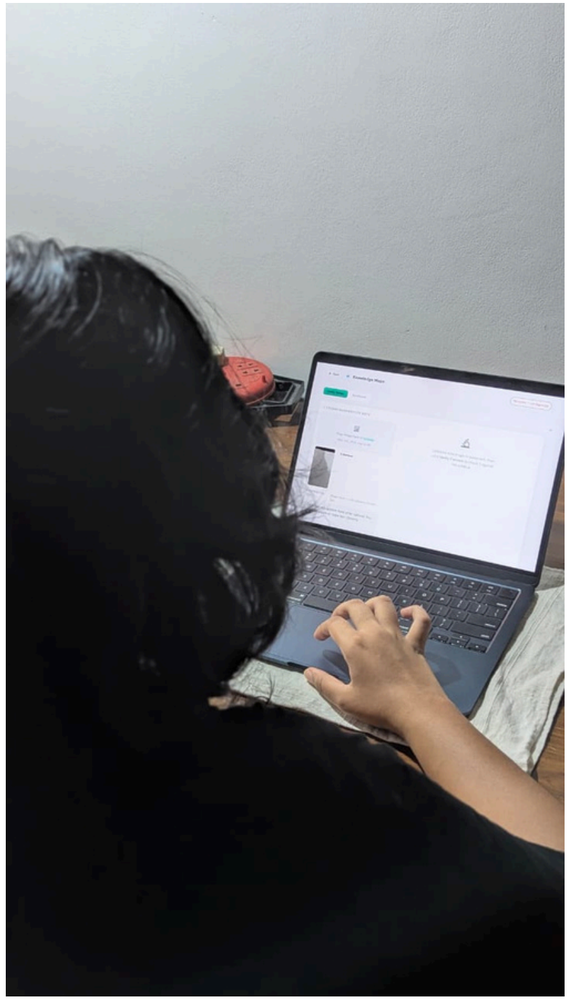
STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

Results and Testimonials





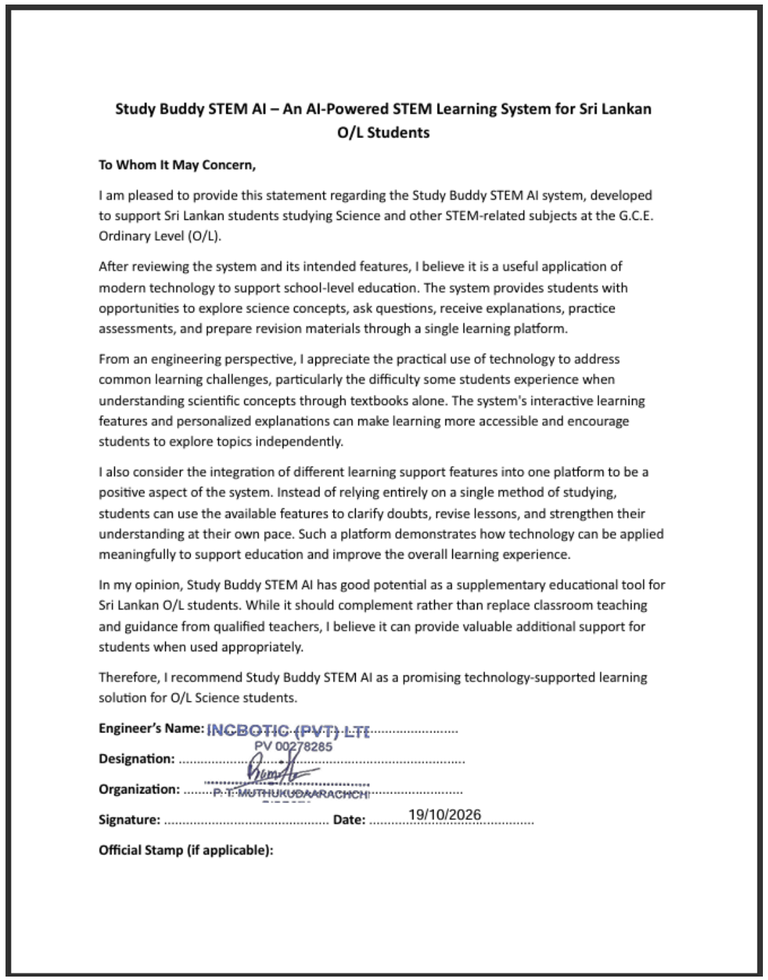




Study Buddy

STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

Results and Testimonials



Study Buddy STEM AI – An AI-Powered STEM Learning System for Sri Lankan O/L Students

To Whom It May Concern,

I am pleased to provide this statement regarding the Study Buddy STEM AI system, developed to support Sri Lankan students studying Science and other STEM-related subjects at the G.C.E. Ordinary Level (O/L).

After reviewing the system and its intended features, I believe it is a useful application of modern technology to support school-level education. The system provides students with opportunities to explore science concepts, ask questions, receive explanations, practice assessments, and prepare revision materials through a single learning platform.

From an engineering perspective, I appreciate the practical use of technology to address common learning challenges, particularly the difficulty some students experience when understanding scientific concepts through textbooks alone. The system's interactive learning features and personalized explanations can make learning more accessible and encourage students to explore topics independently.

I also consider the integration of different learning support features into one platform to be a positive aspect of the system. Instead of relying entirely on a single method of studying, students can use the available features to clarify doubts, revise lessons, and strengthen their understanding at their own pace. Such a platform demonstrates how technology can be applied meaningfully to support education and improve the overall learning experience.

In my opinion, Study Buddy STEM AI has good potential as a supplementary educational tool for Sri Lankan O/L students. While it should complement rather than replace classroom teaching and guidance from qualified teachers, I believe it can provide valuable additional support for students when used appropriately.

Therefore, I recommend Study Buddy STEM AI as a promising technology-supported learning solution for O/L Science students.

Engineer's Name: INCBOTIC (PVT) LTD ..... PV 00278285

Designation: .....

Organization: ..... P.T. MUTHUKUDAARACHCHI .....

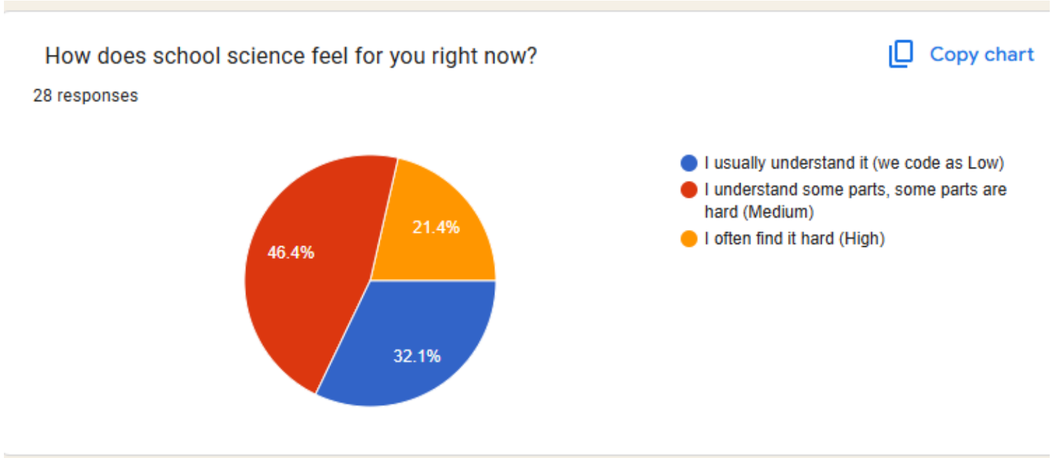
Signature: ..... **Date:** 19/10/2026

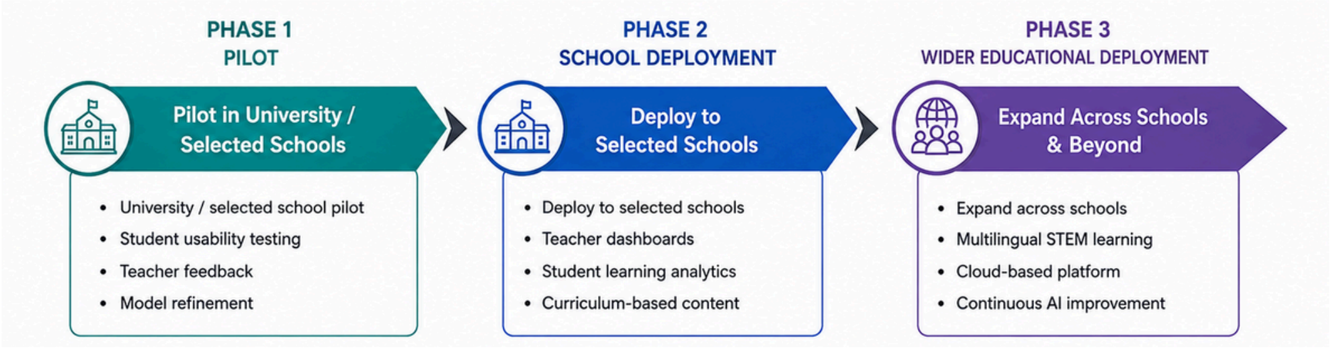
Official Stamp (if applicable):

Study Buddy

STEM AI Integrated AI-Powered STEM Learning Ecosystem for Personalized, Adaptive & Multimodal Education

ABILITY OF COMMERCIALIZATION





Start Small. Validate Learning Impact. Scale Across Sri Lankan Education.

THANK YOU…